

Notes: Jones & Romer (AEJ: Macroeconomics, 2010)

The New Kaldor Facts: Ideas, Institutions, Population, and Human Capital

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1 Big picture

- **Question.** What stylized facts should modern growth theory explain, now that the neoclassical model has largely absorbed Kaldor's original facts?
- **Main move.** Jones and Romer update Kaldor's 1961 list. Instead of focusing mainly on physical capital, the new facts force growth theory to center on four state variables: **ideas, institutions, population, and human capital**.
- **New facts.** The paper proposes six new stylized facts: rising market extent, accelerating growth, greater variation in growth far from the frontier, large income and TFP differences, rising human capital, and stable long-run relative wages.
- **Core mechanism.** The defining feature of ideas is **nonrivalry**: the same idea can be used by many people at the same time. Nonrivalry generates scale effects, increasing returns, incentives for integration, and a role for institutions.
- **Why this matters.** Growth theory cannot be organized around physical capital alone. It must account for how ideas are created and shared, how institutions shape access to ideas, how population affects idea discovery, and how human capital complements technological change.
- **Role of institutions.** Institutions matter because they determine whether ideas are produced, diffused, and used. Poor institutions can block both innovation and adoption of ideas from elsewhere.
- **Human capital theme.** The paper recasts Kaldor's physical-capital facts in human-capital terms: schooling rises strongly, but wage premia for skilled labor do not trend downward, suggesting skill-biased technical change.
- **Economic takeaway.** Modern growth is a general equilibrium interaction between nonrival ideas, scale, human capital, population, and institutions. The paper is not an empirical causal paper; it is a roadmap for what unified growth models must explain.

2 Incremental contribution of the paper

- **From Kaldor's facts to modern growth facts.** The paper's first contribution is to redo Kaldor's exercise for the post-endogenous-growth era. It shows that the core empirical agenda has moved beyond capital accumulation and toward ideas, institutions, demography, and human capital.
- **Unifies several research programs.** The paper links literatures that are often studied separately: endogenous growth, trade and market size, development accounting, institutions, population dynamics, human capital accumulation, and directed technical change.
- **Clarifies why nonrivalry is central.** The paper emphasizes that nonrivalry has two distinct implications: scale effects and institutional design. This distinguishes the scale logic of ideas from the intellectual-property problem of incentivizing innovation.
- **Reinterprets globalization.** The paper argues that rising trade, FDI, patent flows, urbanization, and the web are not just side facts; they are manifestations of the same underlying pressure to expand the market over which ideas can be shared.
- **Connects long-run history to modern growth.** By tying population and idea production together, the paper connects Malthusian history, industrialization, and contemporary growth into one framework.
- **Reframes development gaps.** The paper moves away from pure capital-deepening explanations of cross-country income gaps. It argues that institutions and technology adoption are central to both the level of TFP and catch-up growth.
- **Gives a research agenda.** The paper does not present a new estimated model. Its incremental contribution is agenda-setting: it states the facts a modern unified growth model must fit and explains why the variables interact.

3 Data and measurement behind the stylized facts

3.1 Extent of the market

- The paper uses world trade as a share of world GDP and world FDI as a share of world GDP to document globalization.
- Trade captures flows of goods; FDI captures deeper production and ownership integration.
- The discussion also includes flows of ideas and people, such as patent statistics, urbanization, and information networks.

Interpretation: The figure is not meant to be a full measure of market extent. It is a visual proxy for a broader phenomenon: more people can access and reuse more ideas across larger networks.

3.2 Long-run growth and population

- The long-run data use Maddison-style series for population and per capita GDP for the West.
- Both series are normalized to 1.0 at the initial date and plotted on a log scale.
- The log scale highlights acceleration: the slope itself increases over time.

Interpretation: The point is not merely that levels rise. It is that growth rates rise. This fact motivates models in which ideas and population reinforce one another.

3.3 Growth variation and the technology frontier

- Figure 3 uses Penn World Tables data to plot 1960-2000 growth against 1960 income relative to the United States.
- Rich frontier countries cluster around stable growth rates.
- Poorer countries display much wider dispersion, including rapid catch-up and severe stagnation.

Interpretation: Distance from the frontier creates potential for fast catch-up, but only if institutions and adoption channels allow countries to use frontier ideas.

3.4 Income and TFP differences

- Figure 4 compares per capita GDP and total factor productivity in 2000, normalized so that the United States equals one.
- The poor countries are not only capital-poor; they are also far below the frontier in measured TFP.
- The TFP accounting follows the development-accounting tradition associated with Hall and Jones.

Interpretation: Measured inputs explain less than half of income differences. A large residual remains, and the paper interprets this residual through institutions and idea adoption.

3.5 Human capital and relative wages

- Figure 5 uses U.S. schooling by birth cohort from Goldin and Katz.
- Figure 6 uses U.S. college and high-school wage premia.
- The key juxtaposition is that schooling rises strongly, while skill premia do not show a sustained decline.

Interpretation: The increase in skilled-labor supply did not push down the skill price. A natural explanation is that technical change shifted demand toward educated labor.

4 Models and theoretical specifications

4.1 Production with rival and nonrival goods

The paper starts from a production possibility representation in which output depends on a stock of ideas, rival inputs, and labor:

$$Y = F(A, X, L), \quad (1)$$

where A is nonrival ideas, X is rival resources or physical goods, and L is labor.

The replication argument for rival goods implies constant returns in X and L conditional on A , so output per person can be written as

$$\frac{Y}{L} = F\left(A, \frac{X}{L}, 1\right). \quad (2)$$

Interpretation: With a fixed stock of ideas, more people reduce rival resources per person. But if the stock of ideas rises with people, larger scale can raise output per person because nonrival ideas can be reused by everyone.

4.2 Scale effect from nonrivalry

A simple way to summarize the scale logic is

$$y = mL^\gamma, \quad (3)$$

where y is output per person, L is the number of people connected to the idea pool, and γ measures the strength of scale benefits.

Interpretation: If China doubles the number of researchers connected to the world idea frontier, countries sharing ideas with China can gain even without conventional goods-trade gains. The benefit comes from nonrivality of ideas.

4.3 Population-ideas feedback model

To explain accelerating growth, the paper presents a simple model in which land is fixed, ideas raise productivity, and people generate ideas:

$$Y_t = A_t X^\beta L_t^{1-\beta}, \quad X = 1, \quad (4)$$

$$\dot{A}_t = \alpha L_t. \quad (5)$$

With the Malthusian condition that per capita output equals subsistence,

$$\frac{Y_t}{L_t} = \bar{y} = 1, \quad (6)$$

population is proportional to a power of ideas. Substituting into idea production yields

$$\dot{A}_t = \alpha A_t^{1/\beta}. \quad (7)$$

Interpretation: Since $\beta < 1$, the exponent $1/\beta$ exceeds one. Idea production accelerates as the stock of ideas grows, and population growth accelerates with it. This explains why very long-run growth rates can rise over time rather than stay constant.

4.4 Institutions and TFP differences

Development accounting can be summarized as

$$y_i = A_i k_i^\alpha h_i^{1-\alpha}, \quad (8)$$

where A_i is a residual productivity term. In the paper's interpretation, low A_i reflects barriers to using ideas, not just missing technology in a narrow engineering sense.

Interpretation: Institutions affect the discovery, importation, and use of ideas. A country can remain poor not only because it lacks capital or education, but because institutions block technology adoption and efficient resource allocation.

4.5 Human capital and directed technical change

A useful way to understand the wage-premium fact is through a relative demand representation:

$$\log \left(\frac{w_H}{w_L} \right) = \log \left(\frac{A_H}{A_L} \right) - \frac{1}{\sigma} \log \left(\frac{H}{L} \right), \quad (9)$$

where H/L is relative skilled labor supply and A_H/A_L summarizes skill-biased technology.

Interpretation: If schooling rises but the skill premium does not fall, then demand for skill must also be shifting out. Nonrival ideas can generate directed technical change because innovators target technologies to large markets of users.

5 Theoretical design and credibility of the argument

5.1 What makes the paper persuasive

- The six facts are broad, visually transparent, and connected to large literatures.
- The facts do not require exact econometric identification to be useful; they are stylized constraints on growth theory.
- The paper's strength is conceptual unification: it explains why facts about globalization, human capital, population, and development accounting belong in one model.

5.2 What the paper does not do

- It does not estimate a new structural growth model.
- It does not claim causal identification for each figure.
- It does not provide a full theory of institutional dynamics.
- It does not resolve the optimal institutional design for nonrival ideas.

5.3 Why the approach still matters

The paper is deliberately agenda-setting. Its contribution is to discipline model building: a modern growth model should be judged by whether it can jointly account for these facts rather than explaining each in isolation.

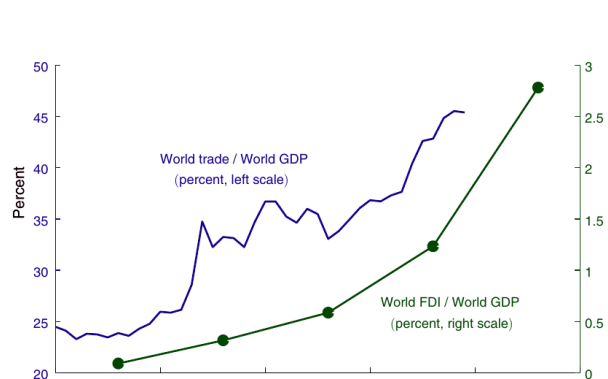
6 Tables and figures

6.1 Figure 1: The Rise in Globalization + Figure 2: Population and Per Capita GDP over the Very Long Run

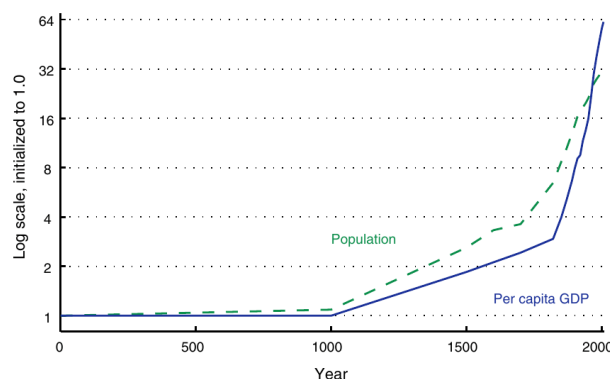
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- **Figure 1** shows world trade as a share of world GDP almost doubling from the 1960s to the late 1990s, while FDI as a share of world GDP rises even more sharply.
- The two series capture different forms of integration: trade in goods and cross-border ownership/production.
- **Figure 2** shows population and per capita GDP in the West over roughly 2,000 years on a log scale.
- The slope of both series increases over time, meaning that growth rates themselves accelerate.
- The figures jointly connect market extent and growth acceleration: bigger integrated populations create larger markets for ideas.

Economic message: Figure 1 documents the expanding domain over which ideas and goods circulate. Figure 2 shows why ideas and population create a historical feedback loop: more people can generate more ideas, and more ideas can support more people and higher incomes.



(a) Figure 1. The Rise in Globalization



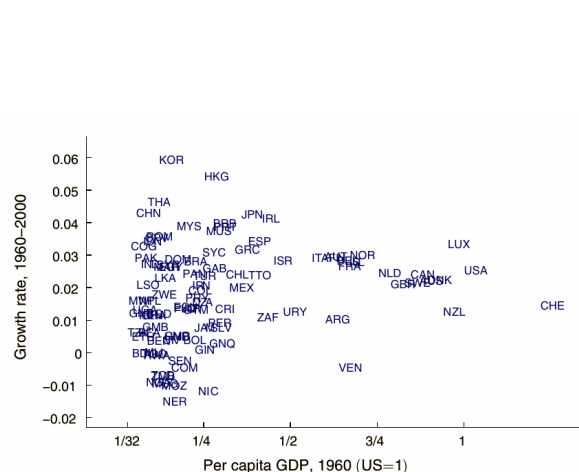
(b) Figure 2. Population and Per Capita GDP over the Very Long Run

6.2 Figure 3: Growth Variation and Distance from the Frontier + Figure 4: Large Income and TFP Differences

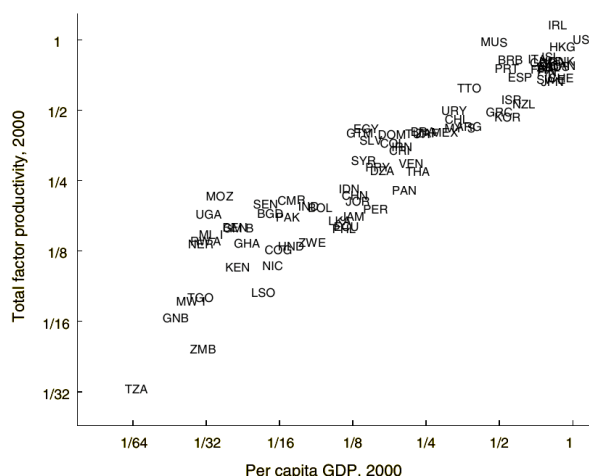
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- **Figure 3** plots 1960-2000 growth rates against 1960 per capita GDP relative to the United States.
- Rich countries near the frontier have relatively similar growth rates.
- Poorer countries show a much wider range: some catch up rapidly, while others stagnate or decline.
- **Figure 4** plots TFP against per capita GDP in 2000, both normalized to U.S. levels.
- Income differences are strongly correlated with TFP differences; poor countries are not just low-capital countries.

Economic message: Figure 3 says distance from the frontier creates potential for fast growth, but Figure 4 says institutions and TFP determine whether that potential is realized. Together they shift the focus from capital accumulation to technology adoption and institutional quality.



(a) Figure 3. Growth Variation and Distance from the Frontier



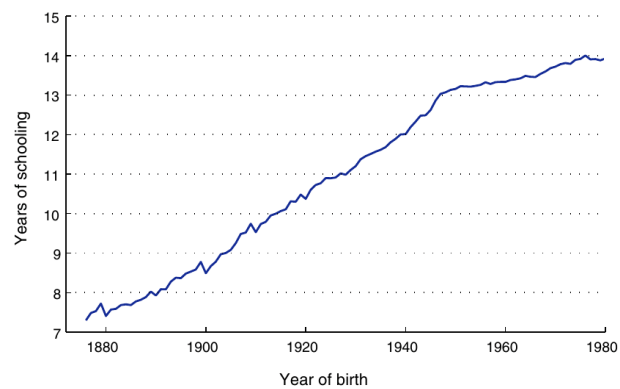
(b) Figure 4. Large Income and TFP Differences

6.3 Figure 5: Years of Schooling by Birth Cohort, United States + Figure 6: The US College and High School Wage Premiums

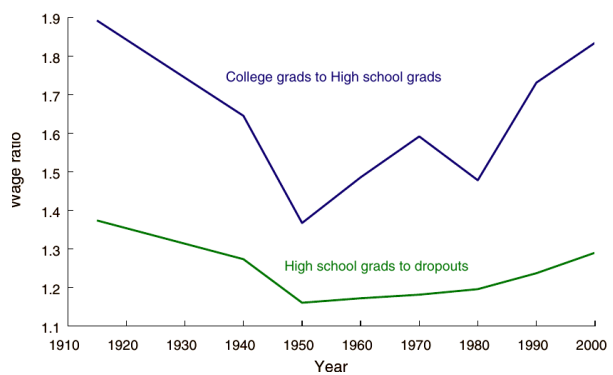
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- **Figure 5** shows U.S. educational attainment rising steadily by birth cohort, from roughly 7-8 years for cohorts born in the late nineteenth century to about 14 years for cohorts born around 1980.
- The later flattening suggests that the contribution of rising schooling to future U.S. growth may slow.
- **Figure 6** shows that college and high-school wage premia fluctuate but do not exhibit a sustained long-run decline.
- The combination is striking: the relative supply of educated labor rises, yet its relative price does not systematically fall.

Economic message: The natural interpretation is skill-biased technical change. Human capital and ideas are complements: more educated workers raise the return to technologies that use skill, and technological change preserves the wage premium despite large increases in schooling.



(a) Figure 5. Years of Schooling by Birth Cohort, United States



(b) Figure 6. The US College and High School Wage Premiums

7 Short summary

- Jones and Romer update Kaldor’s original list of growth facts for modern growth theory.
- Kaldor’s first five facts are now textbook implications of the neoclassical model.
- The new facts require models with ideas, institutions, population, and human capital.
- The first fact is the rising extent of the market, visible in trade, FDI, patent flows, urbanization, and communication networks.
- The second fact is accelerating growth in population and per capita income over the very long run.
- The third fact is that growth-rate dispersion increases with distance from the frontier.
- The fourth fact is that income and TFP differences across countries are enormous and tightly related.
- The fifth fact is rising human capital per worker.
- The sixth fact is stable relative wages despite rising relative human-capital supply.
- Nonrival ideas create scale effects and make institutions central to growth.
- Institutions affect both innovation and adoption of ideas from elsewhere.
- Population and ideas create a feedback loop that can explain accelerating long-run growth.
- Human capital both uses and produces ideas, and skill-biased technical change can explain stable wage premia.
- The paper’s role is not to estimate a new model; it states what a modern unified growth model must explain.

8 Conclusion

8.1 Core conclusion

Jones and Romer argue that the frontier of growth theory has shifted from physical capital toward ideas, institutions, population, and human capital. The neoclassical growth model remains a landmark because it unified Kaldor’s original facts, but modern facts require a broader framework with nonrival ideas and institutional variation at its core.

8.2 Economic intuition

The central intuition is that ideas differ from ordinary goods. A machine, a parcel of land, or a worker’s time can be used in one place at a time. An idea can be used everywhere simultaneously. This nonrivalry gives scale a special role: the larger the connected market, the larger the potential return to ideas. It also means that institutions must balance efficient use of existing ideas with incentives to create new ones.

8.3 Why the six facts belong together

The facts are not a list of unrelated observations. Globalization expands the market for ideas. A larger population can generate more ideas. Institutions determine whether countries can adopt and use ideas. Human capital is both an input into idea production and a complement to new technologies. Stable skill premia despite rising schooling then reflect directed technical change and the growing demand for human capital.

8.4 Incremental lesson for growth theory

The paper's incremental lesson is methodological: growth theory should once again aspire to a unified general equilibrium framework, but the state variables must change. A modern model should jointly explain globalization, accelerating growth, development gaps, technology adoption, schooling, and relative wages. A separate model for each fact would miss the complementarities that make the facts informative.

8.5 Policy interpretation

The paper suggests that policies and institutions affecting idea flows may have very large welfare consequences. Openness, urbanization, education, intellectual property, research universities, and institutional reforms affect not only local efficiency but the global production and diffusion of nonrival ideas. Conversely, institutions that block adoption of ideas can create persistent income gaps.

8.6 Limitations and open questions

The authors acknowledge that institutional dynamics are not yet modeled as cleanly as population, human capital, and ideas. A complete growth theory would explain why good institutions arise, why inefficient institutions persist, and how institutional ideas themselves diffuse. The paper therefore ends with a research agenda rather than a finished model.

8.7 Takeaway

The lasting message is that modern economic growth is an ideas-and-institutions phenomenon. Physical capital still matters, but it is no longer the organizing state variable. The deepest forces are the nonrivalry of ideas, the scale of the market over which ideas can be used, the human capital that creates and applies them, and the institutions that permit or prevent their diffusion.